

		Parul University Faculty of Engineering and Technology Parul Institute of Engineering and Technology Department: AI-ML/AI-RO/AI/AI-DS/CSE/MICRO/SAP/QUICK/ORACLE/IT/AERO		
Subject Name	PROBABILITY, STATISTICS AND NUMERICAL METHODS	A.Y	2025/2026	
Subject Code	303191251	Semester	4th	
Tutorial-3				
Sr No	Question	COs	B.T	Competence
1	A sample of 100 bulbs has a mean life of 1200 hours with standard deviation 90 hours. Test at 5% level whether the mean life is 1250 hours.	3	4	Analyze
2	In a sample of 400 men, 220 are tea drinkers. Can we conclude at 1% level of significance that tea drinking is preferred by majority?	3	3	Apply
3	Two samples of sizes 50 and 60 have means 45 and 48 with standard deviations 8 and 7 respectively. Test the hypothesis that the population means are equal at 5% level.	3	4	Analyze
4	A machine produces items with mean weight 15 g and standard deviation 2 g. A sample of 36 items gives mean weight 15.7 g. Test at 1% significance whether the machine is producing overweight items.	3	5	Evaluate
5	A manufacturer claims that only 5% of items are defective. A random sample of 200 items shows 18 defective items. Test the claim at 5% level.	3	4	Analyze
6	The standard deviation of a random sample of 25 items is 6.5. Test at 5% level whether the population standard deviation is 5.	3	3	Apply
7	Two samples have variances 10.2 and 12.8 with sample sizes 20 and 22. Test whether the population variances differ significantly at 5% level.	3	5	Evaluate

8	A die is thrown 120 times and the following frequencies are obtained: 1:18, 2:20, 3:25, 4:16, 5:22, 6:19. Use Chi-square test to check whether the die is fair.	3	6	Create
9	The following data show the distribution of workers according to age and accident record. Use chi-square test to check independence. Age: (Below 30: 10 accidents, 40 no accidents), (30–50: 25 accidents, 35 no accidents), (Above 50: 15 accidents, 25 no accidents).	3	5	Evaluate
10	A study claims that the mean marks of two sections of a class differ. Section A: $n=30$, $\text{mean}=76$, $\text{SD}=8$. Section B: $n=35$, $\text{mean}=72$, $\text{SD}=10$. Test the claim at 5% significance.	3	4	Analyze